

Week 1 Assignment

Each student must write a short report covering the following topics.

Part 1: Data Structures and Algorithms

1. Explain, using short and simple sentences, what **data structures** and **algorithms** are.
2. Explain why learning data structures and algorithms is important.
3. Identify the built-in data structures supported by Python.
4. For each data structure, describe its important operations. You do not need to mention operations that only return the number of stored elements.

Part 2: Performance Comparison Between Python Lists and Sets

Conduct an experiment to compare the performance of Python `list` and `set`.

For each value of

$$M \in \{10^3, 10^4, 10^5\},$$

complete the following steps:

1. Generate a sequence of M random integers. Each integer must be in the range from 1 to 2×10^5 .
2. Store the generated sequence in:
 - a Python `list`;
 - a Python `set`.
3. Measure the time required to create each data structure.
4. Generate $N = 5 \times 10^4$ search queries, with each query value in the range from 1 to 2×10^5 .
5. Perform all N search operations on:
 - the Python `list`;
 - the Python `set`.
6. Measure the total search time for each data structure.

Report Requirements

Present the measured running times in a table. For each value of M , report:

- the time required to create the `list`;
- the time required to create the `set`;
- the total time required to perform N searches in the `list`;
- the total time required to perform N searches in the `set`.

Finally, compare the results and briefly explain why searching in a `set` is generally faster than searching in a `list`.

Fair-comparison requirement: Use the same generated sequence and the same search queries for both the <code>list</code> and the <code>set</code> .
